INSTITUTO VIRTUDES

Applied Research in Virtology

Pyramid of Virtues and Directed Neuroplasticity:

Preliminary empirical evidence from a brief virtological intervention with families in social vulnerability

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Abstract

Objective. To evaluate, in an exploratory quasi-experimental design, the effect of a brief virtological intervention on four virtues in the Virtology matrix - Humility, Gentleness, Patience and Responsibility - among family members of children and adolescents in social vulnerability, and to interpret the results in light of the Pyramid of Virtues (CASAROTTO, 2026) as a clinical organizer of the neuroplastic sequencing of the thirty-three virtues. Method. Six family members linked to a Brazilian non-profit organization participated in the study (intervention group, n=6), compared with eight participants who received only common-sense theoretical exposure (control group, n=8). The intervention, comprising five weekly meetings conducted in January 2026 by three practitioners trained in Virtology, combined accessible language, experiential activities and paired exercises in directed neuroplasticity in accordance with the PRF formula of Casarotto's Law. A twelve-item questionnaire with different weights for response options A-E generated simultaneous scores for twenty-eight virtues in the matrix, administered at pre-test and post-test. Participants signed an Informed Consent Form. Results. The five intervention-group participants who completed the post-test showed average gains of between 32.5% and 37.6% in the four primary virtues. A paradigmatic case (P1) rose from the lowest score in the entire sample to the upper band (+66 points across the primary virtues), a trajectory compatible with the virtological notion of successful Treatment. An anomalous case (P5) showed homogeneous regression (−35 points overall), with all five responses changed at post-test moving toward lower alternatives; item-by-item analysis suggests a gain in self-awareness and a reduction in social desirability bias, a pattern compatible with entry into the Partial Awareness Band described in the Pyramid of Virtues. Gains extended to untrained secondary virtues (Acceptance, Flexibility, Purity of Heart, Positive Example, Love of Neighbor, Fraternity and We Are All ONE), in a pattern of network-based neuroplastic inscription. Conclusion. The intervention produced gains compatible with a leap in Evolution Band among participants in social vulnerability, without producing virtological iatrogenesis, even though it simultaneously crossed the Base Block (Humility, Responsibility) and the Middle Block (Gentleness, Patience) of the Pyramid. The findings offer preliminary empirical evidence for the central thesis of clinical Virtology articulated by Casarotto: virtues are neural structures trainable through directed neuroplasticity, and their sequencing in the Pyramid is the organizer that distinguishes genuine evolution from a superficially performing Persona.

Keywords

Virtology; Pyramid of Virtues; Casarotto's Law; Directed neuroplasticity; Evolution Bands; Human Needs; Tripartite model; Community intervention.

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Keywords

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1. Introduction

Virtology, a line of research that brings together the philosophy of virtues, depth psychology and contemporary neuroscience, advances a precise clinical thesis: virtues are neither abstract moral predicates nor ornaments of personality, but neural structures trainable through directed neuroplasticity, capable of reorganizing, within nervous tissue, the subject's relationship with themselves, with others and with reality (CASAROTTO, 2024; CASAROTTO, 2026). This thesis, supported by extensive theoretical articulation in recent years, nevertheless calls for empirical evidence in naturalistic settings - not only in the laboratory, not only in individual clinical practice, but in community contexts where social vulnerability, material insecurity and everyday psychological suffering constitute the actual ground on which the intervention must operate.

This article presents the results of exploratory research conducted in January 2026 with families served by a Brazilian non-profit organization working with children and adolescents at risk and in social vulnerability. The intervention, brief by design - five weekly meetings conducted by three practitioners trained in Virtology - systematically worked on four virtues in the virtological matrix: Humility, Gentleness, Patience and Responsibility. This choice is not arbitrary. In light of the Pyramid of Virtues proposed by Casarotto (2026) as a clinical organizer for work with the thirty-three virtues in the matrix, these four virtues occupy strategic positions spanning the Base Block (Treatment) and the Middle Block (Development). Humility and Responsibility belong to the Base: they address, respectively, pride - a primitive brainstem vector identified as “the zero point of all human neuroses” - and the outsourcing of agency, two axes whose dismantling is a neuropsychic precondition for any subsequent work. Gentleness and Patience, in turn, belong to the Middle: they are virtues of affective modulation and tolerance for the time of reality, associated with the prefrontal cortex operating as an executive inhibitor of limbic automatisms (ARNSTEN, 2009; OCHSNER; GROSS, 2014).

The empirical question is direct: does a brief intervention, conducted in language accessible to people in social vulnerability, produce detectable neuroplastic inscription in virtues of the virtological matrix? And, if so, is this effect restricted to the virtues directly trained, or does it extend, in a network pattern, to interwoven secondary virtues? More fundamentally: does the sequencing proposed by the Pyramid of Virtues operate as a valid clinical organizer - or does simultaneously crossing the Base and Middle, as this intervention did, produce iatrogenesis, frustration and regression, as predicted by the chapter devoted to the pitfalls of “skipping stages” (CASAROTTO, 2026)?

The contribution of this article is threefold. First, it offers preliminary empirical evidence - preliminary, given the small sample and the methodological limitations that will be discussed candidly - for the thesis that virtues are trainable in community settings, even among populations in vulnerability. Second, it articulates this evidence with the Pyramid of Virtues,

showing that the results are consistent with what the theory predicts about transitions between Evolution Bands and the Human Needs Groups as neurobiological mediators of the effect. Third, it integrates the virtological reading with contemporary neuroscientific literature - particularly Kandel's neuroscience of synaptic plasticity, Damasio's theory of somatic markers, Menon's tripartite model of cortical networks, and the contemplative neuroscience of Davidson, Tang and Lazar - creating a dialogue in which Virtology positions itself as a framework integrating these contributions, rather than as a discourse parallel to them.

The article is organized into seven sections. The next reconstructs the theoretical framework - the Pyramid, Bands, Casarotto's Law and the neuroscience of virtues. The third details the method. The fourth presents quantitative and qualitative results. The fifth discusses the findings in light of the Pyramid and contemporary neuroscience. The sixth candidly sets out the limitations. The seventh concludes with propositions and directions for the next research cycle.

2. Theoretical framework

2.1. The Pyramid of Virtues as a clinical organizer

Virtology organizes the thirty-three virtues in its matrix into three clinically sequenced blocks, described by Casarotto (2026) as the Pyramid of Virtues. This organizer does not answer the neuroscientific question of what each virtue is - the subject of the tripartite model of the DMN, SN and CEN networks - but the clinical question of when and in what order each virtue should be worked on, with which patient, and in which Evolution Band. The Pyramid is, in other words, the neuroplastic sequencing of work with virtues.

The Base of the Pyramid brings together eight virtues - Humility, Selflessness, Acceptance, Forgiveness, Individuation, Responsibility, Faith and Perseverance - whose clinical function is to draw the individual out of primitiveness. Each operates on a specific vector of primitive functioning: Humility dismantles pride, “the zero point of all human neuroses”; Selflessness addresses selfishness; Acceptance addresses lack as resistance to reality; Forgiveness addresses lack in its relational dimension; Individuation addresses dependence on the other's gaze; Responsibility addresses the outsourcing of agency; Faith addresses anxiety and fear of chaos; Perseverance sustains the therapeutic process itself against abandonment. Without work on these eight virtues, no virtue of the higher block can become established as second nature - what is obtained, in the absence of the Base, is Persona performance, not a constructed Self.

The Middle of the Pyramid brings together twenty virtues - Gentleness, Patience, Good Humor, Gratitude, Optimism, Truth, Honesty, Purity of Heart, Non-Judgment, Flexibility, Pursuit of Improvement, Care for the Body, Reconciliation, Repaying Evil with Good, Honoring the Family, Positive Example, Utility, Eternity, Love of Nature and Detachment - whose clinical function is to align the individual with the Laws of the Universe and the conscious management of their needs. Detachment,

in particular, operates as a gateway to the third block, insofar as it detaches the Ego from the material and identity-based supports that sustain its defensive organization.

The Top of the Pyramid brings together five virtues - We Are All ONE, Fraternity, Love of Neighbor, Helping Others and Equality - whose clinical function is to raise the subject to Feeling, to the authentic Self and to lived purpose, transcending the Ego in favor of the universal Self. These five virtues are all relational and altruistic: the movement beyond oneself that is realized in service to others and to the whole.

The immediate clinical consequence of the Pyramid is decisive: no therapist can ask for charity from someone who has not yet constructed an Ego; none can ask for detachment from someone who has not yet emerged from primitiveness; none can ask for transcendence from someone who has not yet developed a personality. Skipping stages of the Pyramid is the most common form of iatrogenesis in therapies inspired by virtues - at best, it produces performative improvement; at worst, chronic frustration and regression of the Band.

> **The pyramid as a map, the sequence as a path**
>
> I set out to work on the base of the structure. If I have not worked on acceptance, have not worked on forgiveness, have not worked on faith, perseverance - have not worked on those foundations in relation to this case - and I stay there in gratitude, I am going to be grateful and light, I am going to skip stages. This is the rule that must be inscribed in every virtological therapist: the work begins at the base of the pyramid. Always. - Dr. Eduardo Casarotto, Class of May 13, 2026 - The Psychic Apparatus.

2.2. Evolution Bands and Needs Groups

The Pyramid operates in one-to-one correspondence with two other central architectures of Virtology: the Evolution Bands and the Needs Groups. The Bands - Primitive, Sensation, Pre-Development, Awareness (Partial and Full), Feeling, Full Love and Full Charity - describe, in sequence, the predominant configuration through which the subject's psychic apparatus reads and responds to reality. In neurobiological terms, the lower bands are dominated by the brainstem and the limbic system; the intermediate bands correspond to the emergence of the prefrontal cortex as the executive protagonist; the higher bands correspond to full tripartite integration between DMN, SN and CEN, with progressive suppression of the Ego's defensive automatisms.

The Needs Groups, in turn, are five: G1 (Primitive - importance, security, competition, resolving lack, trust and credibility); G2 (Sensation and Bonding - belonging, feeling loved, aesthetics, creativity); G3 (Development - knowledge, individuality, accomplishing/bringing into being, self-development, restoration and regeneration); G4 (Reconnection - self-fulfillment, connection with nature, charity, Self, purpose, introspection); G5 (Transcendence - self-transcendence, unity with the whole, fullness and completeness). Each Group, read on a virtological scale from 0 to 10, operates in homeostasis when it remains within the range

from 4 to 7. Below 4 (acute deprivation) or above 7 (pathological excess), the Group hijacks the system and no work on a higher virtue can be sustained.

The triple correlation Pyramid × Bands × Needs is the map that the virtological therapist reads to situate the patient. The Base Block (Treatment) corresponds to the Primitive, Sensation and Pre-Development Bands, and to Needs Groups 1 and 2. The Middle Block (Development) corresponds to the Awareness Band (Partial), and to Groups 3 and 4. The Top Block (Transcendence) corresponds to the Feeling and Full Love and Charity Bands, and to Group 5. This mapping is not a theoretical abstraction: it is the operational basis of Virtology's fundamental clinical principle - inhibit the currently dominant needs group and stimulate the group immediately above it, paired with the corresponding virtues from the appropriate block of the Pyramid.

2.3. Casarotto's Law and the PRF formula

Casarotto's Law states that every Perception-Reaction (PR) and the Flow (F) arising from it are a multiplicative function of five main variables: the subjective Intensity of the stimulus (I), the environmental Stimulus (E), the Evolution Band in which the subject operates (FA), Personality with its virtues and driving forces (PS), and the degree of alignment with the Laws of the Universe (LU). The PRF formula takes the form PRF = (I × E) × FA × PO × PS × LU, where PO represents the specific driving forces. The multiplicative nature of the formula is decisive: each variable potentiates the others, and working on one without the others means multiplying by a disproportionately low number, leaving the final product - existential coherence - short of its potential.

For the empirical reading of this study, three variables in the formula are especially relevant. The variable PS is the clinical operationalization of “personality” as a set of trainable competencies - it is within it that virtues become neuroplastically inscribed. The variable FA describes the band through which reality is read, and its progression over time is what is called a “leap in Evolution Band.” The variable I, finally, describes the subjective intensity with which the same environmental stimulus (constant E) is apprehended - a variable that, in clinical practice, is strongly modulated by the three major cortical networks. In operational terms: when, faced with the same E (the daughter who does not obey, the grandson who has a crisis, life that does not follow the script), the subject reports reduced I, more virtuous PS and higher FA, this meets the precise operational definition of a Band gain in the virtological architecture.

> ***PRF = (I × E) × FA × PO × PS × LU - existential coherence as a multiplicative product. Each variable worked on in isolation multiplies by a disproportionately low value. “Everything aligned” - virtue × need × mandate***
>
> ***× band - is the condition for effective neuroplasticity.***

2.4. Virtues as neural structures: a dialogue with contemporary neuroscience

The thesis that virtues are trainable neural structures is not a mere analogy: it rests on five lines of evidence from contemporary neuroscience which, although developed in independent research programs, converge toward a coherent framework when integrated by Virtology.

Synaptic plasticity and the formation of procedural memories

Eric Kandel demonstrated, over five decades, that every subjective experience - including the clinical structures described by psychoanalysis - is anchored in lasting synaptic modifications, and that these modifications are, in principle, reversible through new experiences (KANDEL, 1998; KANDEL, 2006). The disciplined repetition of a pattern of thought, emotion and action produces, through molecular processes, alterations in the strength of synaptic connections that, over weeks and months, consolidate into procedural memory traces - precisely the neural substrate Virtology describes when it states that a virtue "operates as second nature." Kandel is especially important to the present articulation because, unlike most contemporary neuroscientists, he never abandoned the Freudian tradition - his thesis is that contemporary neuroscience is, in a strong sense, the heir to psychoanalysis, and that neurosis, psychosis and perversion are consolidated synaptic patterns with varying degrees of residual plasticity available for clinical work (KANDEL, 2018).

Somatic markers and body-emotion-cognition integration

Antônio Damasio, through the somatic marker hypothesis (DAMASIO, 1994; DAMASIO, 2010), demonstrated that apparently "rational" decisions are, in reality, anchored in somatic signals rapidly assessed by the ventromedial prefrontal cortex (vmPFC). The separation between cognition and emotion - presupposed by Cartesian models - is, in contemporary neuroscience, biologically untenable. For Virtology, this means that work with virtues is not disembodied "cognitive" work, but work that necessarily recruits the body, the interoceptive system and corticolimbic integration: training Gentleness is not deciding to be gentle - it is reorganizing the amygdala-vmPFC circuit so that the somatic marker "offense = explode" is replaced by the marker "offense = pause, choice, modulation."

The tripartite model of cortical networks

Vinod Menon and Michael Bressler proposed, in a synthesis now widely accepted in cognitive neuroscience, that three major cortical networks regulate most higher mental processing: the Default Mode Network (DMN), responsible for self-reference, autobiographical narrative and theory of mind; the Salience Network (SN), centered on the insular cortex and the anterior cingulate cortex, responsible for registering what matters; and the Central Executive Network (CEN), centered on the dorsolateral prefrontal cortex, responsible for

executive cognition and deliberate action (MENON, 2011; BRESSLER; MENON, 2010). Virtology, in the chapter devoted to the tripartite model, articulates each virtue as a specific signature of transition between these three networks - a description that does not replace the clinical description of the Pyramid, but complements it at the neurobiological level.

Allostatic stress and prefrontal function

Amy Arnsten demonstrated that the prefrontal cortex is, paradoxically, the structure most vulnerable to the effects of chronic stress: under a high allostatic load, the noradrenergic system progressively disorganizes prefrontal connections, leaving the limbic system - especially the amygdala - hyperreactive (ARNSTEN, 2009; ARNSTEN, 2015). Bruce McEwen and Robert Sapolsky described, in parallel, how chronic exposure to cortisol remodels the hippocampus, reduces prefrontal dendritic plasticity and crystallizes patterns of hypervigilance (MCEWEN, 2017; SAPOLSKY, 2017). For Virtology, this convergence is decisive: populations in social vulnerability - precisely the population of this study - typically operate under a high allostatic load, with a chronically compromised prefrontal cortex. The virtues of the Base of the Pyramid are, in this sense, a precise intervention in this circuit: they restore prefrontal inhibitory function over the amygdala, and do so through behavioral, not pharmacological, means.

Contemplative training and structural plasticity

Sara Lazar demonstrated, in seminal studies, that regular meditation practitioners show increased cortical thickness in prefrontal regions and in the insular cortex (LAZAR et al., 2005). Britta Hölzel extended this evidence by showing structural changes detectable after only eight weeks of MBSR training (HÖLZEL et al., 2011). Yi-Yuan Tang and Michael Posner developed Integrative Body-Mind Training (IBMT), demonstrating gains in executive control and affective modulation after a few days of training (TANG; HÖLZEL; POSNER, 2015). Richard Davidson, over decades of research in contemplative neuroscience, demonstrated that emotional styles are plastic and trainable throughout adult life (DAVIDSON; MCEWEN, 2012; DAVIDSON; BEGLEY, 2012). Tania Singer and Olga Klimecki, in studies on compassion training, demonstrated that even "relational" virtues such as Fraternity and Love of Neighbor recruit specific circuits - the care and reward system - and are trainable through structured exercises (SINGER; KLIMECKI, 2014; KLIMECKI et al., 2013). This literature provides the most direct neuroscientific support for the virtological thesis that virtues are not innate moral predicates, but competencies trainable at any stage of the life cycle.

Virtology integrates these five lines into a single operational framework. Virtues are neuroplastic patterns sustained by lasting synaptic modifications (Kandel), anchored in integrated somatic markers (Damasio), expressed as signatures of transition between the DMN-SN-CEN networks (Menon), capable of restoring prefrontal function chronically compromised by allostatic stress (Arnsten, McEwen, Sapolsky), and demonstrated to be trainable in diverse populations by the literature on contemplative neuroscience (Lazar, Hölzel,

Tang, Posner, Davidson, Singer, Klimecki). The Pyramid of Virtues is the clinical organizer that orders this work over the patient's evolutionary time - not as a menu of desirable competencies, but as a neuropsychically structured sequence.

3. Method

3.1. Institutional context

The research was conducted in partnership with a Brazilian non-profit NGO called Nova Geração Brasil, located in São João del Rei, a municipality in the interior of the state of Minas Gerais. The philanthropic organization serves children and adolescents aged six to eighteen who are at risk and in social vulnerability, offering activities outside school hours aimed at personal and social development, affirmation of identity, citizenship and self-esteem. The intervention was offered to family members (mothers, grandmothers and other guardians) of the children served, on the basis of voluntary participation. A significant proportion of the children have neurodevelopmental disorders, a context that makes work with family members particularly relevant - these adults are, in everyday life, the main affective regulators of these children, and their evolution band has a direct impact on the children's developmental environment.

3.2. Participants

The sample consisted of fourteen adults, all family members of children and adolescents served by the organization. Participants were allocated, on the basis of their availability and voluntary adherence to the process, into two groups: an intervention group (n=6) and a control group (n=8). The general sociodemographic profile includes predominantly basic schooling, low family income and family configurations marked by a heavy caregiving burden. To respect the confidentiality of the research participants, identified by alphanumeric codes (P1 to P6 for intervention; C1 to C8 for control), no personal data are presented in this article. All participants signed an Informed Consent Form (TCLE) and were informed about the conditions of the research, the voluntary nature of participation and the anonymous use of their data.

3.3. Design

An exploratory quasi-experimental design with pre-test, post-test and a control group was adopted. The intervention group received five weekly meetings of approximately ninety minutes, conducted by three volunteer practitioners from the organization - professionals who had completed or were undertaking training in Virtology. During the same period, the control group received only common-sense theoretical exposure to the topics, without the structure of systematic practice, experiential activities or paired exercises in directed neuroplasticity that characterize the virtological intervention itself. For operational reasons

related to the functioning of the organization, the control group did not complete the post-test - a relevant methodological limitation that will be discussed in Section 6.

3.4. Intervention

The intervention brought together four virtues in the virtological matrix - Humility, Gentleness, Patience and Responsibility - in a structure inspired by the protocols of clinical Virtology and adapted to the language and time available to the population served. Each meeting addressed one of the virtues in depth, combining: initial investigation (mapping each participant's Evolution Band through accounts of everyday life); teaching (a technical explanation, in accessible language, of the relationship between the virtue being worked on and its corresponding neurotic vector); action (exercises in directed neuroplasticity, in fourteen-day cycles, paired with real situations in each participant's everyday life); and synthesis (consolidation at the end of the session, with a task between meetings). The main teaching material was Guia das Virtudes, chosen specifically for its accessible language and its practical connection with concrete scenes from everyday life. The fifth meeting served as a consolidation and closing session, with administration of the post-test.

3.5. Instrument

The quantitative measure used a twelve-item questionnaire - three items for each of the four primary virtues - consisting of everyday-life situations with five response options (A-E). Each option received different weights, calibrated to generate simultaneous scores for up to sixteen secondary virtues in the matrix, in addition to the item's primary virtue. Twenty-eight virtues were scored (five virtues in the matrix - Honesty, Eternity, Detachment, Care for the Body, and Love of Nature and Animals - were not assigned weights in the present instrument and were excluded from the analysis). The questionnaire was administered in structured meetings, with practitioners reading the questions individually to participants when necessary, given the sample's heterogeneous literacy levels.

Instrument characteristic	Specification
Number of items	12 (3 per primary virtue)
Response structure	5 options A-E with different weights
Virtues scored	28 (4 primary + 24 secondary)
Administration format	Individual, with assisted reading when necessary
Average administration time	20 to 30 minutes
Administration points	Pre-intervention and post-intervention (4 weeks)

Table 1. Specifications of the measurement instrument.

3.6. Data analysis

The quantitative analysis was conducted on each participant's raw scores for each of the twenty-eight scorable virtues at both time points. The following were calculated: (i) means by group and virtue, (ii) absolute and relative deltas (post-pre), (iii) individual trajectories in the four primary virtues, and (iv) patterns of gains in secondary virtues. For the anomalous case (P5), an item-by-item analysis compared the options selected at pre-test and post-test to identify the directional pattern of changes. The qualitative interpretation was triangulated with the practitioners' report and three spontaneous testimonials collected at the end of the intervention, grounding the quantitative findings in the participants' own phenomenological descriptions.

3.7. Ethical considerations

All participants signed an Informed Consent Form after receiving a detailed explanation of the nature of the research, the procedures involved, the anonymous use of the data and the right to withdraw from the study at any time without prejudice to the services provided by the organization. Participants were informed about the exploratory nature of the research and the absence of foreseeable risk greater than that inherent in everyday exposure to the topic of virtues. To respect confidentiality, no personal data - name, age, specific family configuration, medical condition or individualized institutional connection - are presented in this article, with participants identified exclusively by alphanumeric codes.

4. Results

4.1. Baseline characterization

The first relevant finding is that, at pre-test, the intervention group started with a lower mean score than the control group in all four primary virtues. The differences were particularly marked in Gentleness (−2.13 points) and Patience (−2.60 points) - precisely the two virtues most closely associated, within the virtological architecture, with affective modulation and tolerance for the interval between trigger and response. The differences in Humility (−0.90) and Responsibility (−0.97) were smaller, but in the same direction (Figure 1).

Figure 1. Baseline: mean scores in the four primary virtues by group, at pre-test

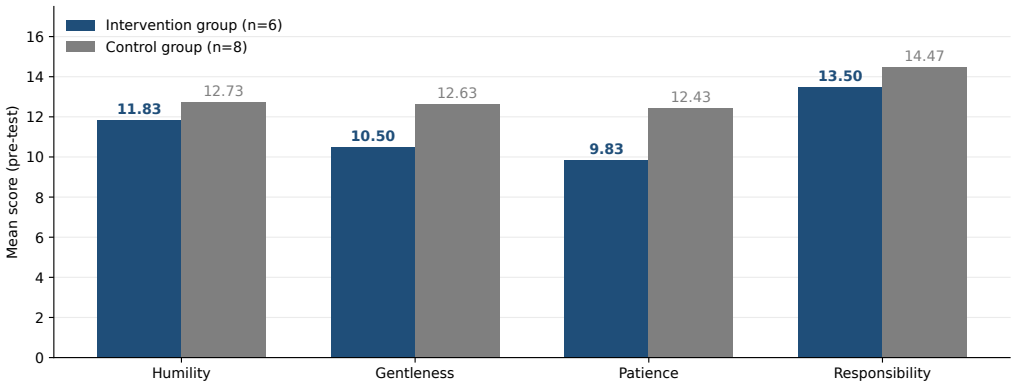


Figure 1. Mean pre-test scores for the four primary virtues by experimental condition. The intervention group started at a lower level on all four measures.

The clinical relevance of the lower baseline

This finding has theoretical importance beyond the metric. The intervention was applied to the group in the less favorable condition, rather than to the group with better prior performance. In terms of Casarotto's Law, this means that the intervention group started with weaker PS and lower FA - precisely the profile for which the Base Block of the Pyramid of Virtues is clinically indicated (CASAROTTO, 2026). In terms of the literature on chronic allostatic stress, the profile is compatible with nervous systems with a chronically overloaded prefrontal cortex and a hyperreactive amygdala (ARNSTEN, 2009; SAPOLSKY, 2017), making subsequent gains, when they occur, even more pronounced from a neurobiological perspective.

4.2. Mean evolution of the intervention group

Considering the five participants who completed pre-test and post-test, the mean gains in the four primary virtues were substantial and proportionally homogeneous: Humility +37.6%; Gentleness +34.1%; Patience +32.5%; Responsibility +32.9%. The proportional homogeneity of the gains, at around one third of the previous score, is itself a relevant theoretical finding: it suggests a reorganization of the Band as a whole - that is, a shift in the architecture through which reality is read - rather than only a specific gain in a particular competency (Figure 2).

Figure 2. Intervention group progression in the four primary virtues

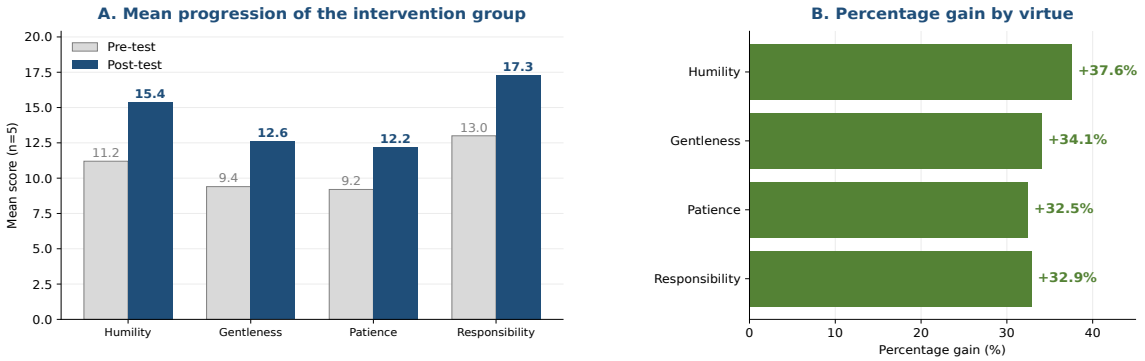


Figure 2. (A) Pre→post-test change in the intervention group across the four primary virtues. (B) Percentage gain per virtue.

4.3. Individual trajectories

The participant-level analysis shows that the group mean consists of quite heterogeneous trajectories. Four participants (P1, P2, P3, P4) showed gains of varying magnitudes. One participant (P5) showed a uniform decline-an anomalous case whose analysis will be presented in detail in Subsection 4.5. Participant P6 did not complete the post-test and was excluded from the pre/post analysis (Figure 3).

Figure 3. Individual trajectories - total of the four primary virtues (intervention group, participants completing pre- and post-test)

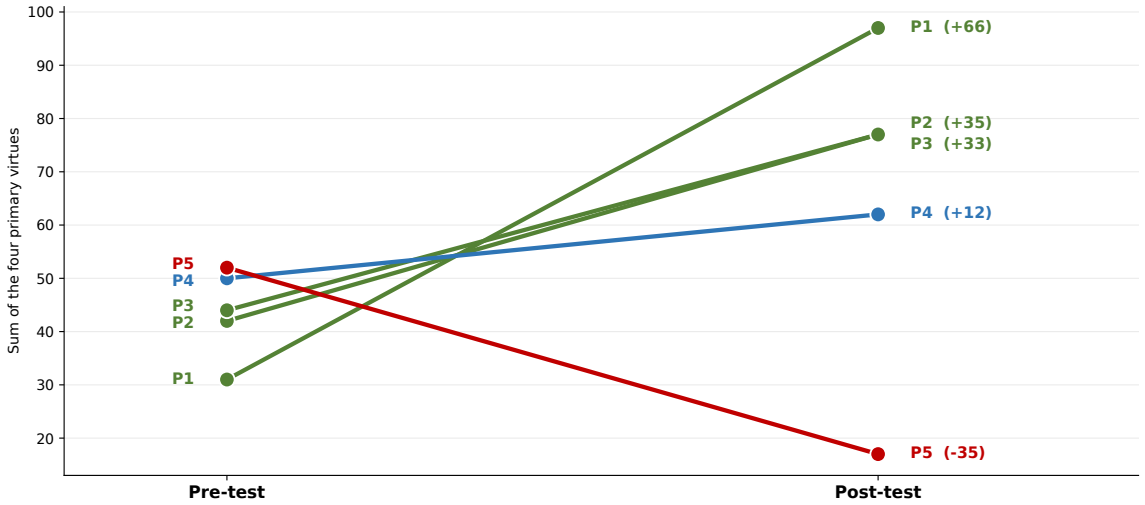


Figure 3. Individual trajectories-the sum of the four primary virtues-for each intervention-group participant who completed the pre- and post-test.

Considering the total for the four primary virtues, the individual deltas were: P1 (+66 points, rising from 31 to 97); P2 (+35 points, from 42 to 77); P3 (+33 points, from 44 to 77); P4 (+12 points, from 50 to 62); P5 (−35 points, from 52 to 17). The virtological analysis differentiates

these trajectories by their clinical significance. P2 and P3 show consistent, balanced improvement in all four virtues simultaneously-a pattern compatible with a reorganization of the Band as a whole, rather than in only one specific dimension. P4 shows real but more modest improvement-probably a combination of a partial ceiling effect (the participant already started with a reasonable score) and life-stage characteristics that warrant individual investigation. P1 is the paradigmatic case, analyzed in the next subsection. P5 is the anomalous case, analyzed afterward.

4.4. Paradigmatic case (P1)

Participant P1 is the most illustrative clinical case in the set. At pre-test, this participant started with the lowest scores observed in the entire sample-Humility 7.0; Patience 6.0; Gentleness 8.0; Responsibility 10.0-a profile compatible with the Primitive Band described in the chapter on the Pyramid of Virtues (CASAROTTO, 2026), in which the psychic apparatus operates predominantly through pride, lack, and dependence on the other's gaze. At post-test, P1 moved to scores close to the highest in the group (Humility 30.0; Gentleness 26.0; Patience 22.0; Responsibility 19.0), a cumulative trajectory of +66 points in the total for the four primary virtues, more than twice the group mean.

In terms of Casarotto's Law, case P1 is compatible with a simultaneous, coordinated increase in I (subjective intensity begins to register nuances that were previously numbed), PS (virtues inscribed as new competencies), and FA (the band rises from primitive dominance toward Partial Awareness configurations). The proportional delta-almost half the total scale in five weeks-is greater than could be explained solely by social desirability bias or familiarity with the instrument, supporting a clinical interpretation of a genuine Band leap. In terms of the Pyramid, P1 is the clearest example of the Base Block working as it should-Humility dismantles pride, thereby opening the way for Gentleness, Patience, and Responsibility to find expression.

4.5. Anomalous case (P5)

Participant P5 showed the opposite pattern: a uniform decline in all four primary virtues, with deltas of −9, −9, −9, and −8 points respectively, and a total delta of −35 points. At first glance, the finding appears to suggest moral regression or failure of the intervention. However, an item-by-item analysis of the instrument's twelve questions revealed a striking directional pattern: all five responses changed at post-test moved to lower alternatives (E→D, E→C, D→C, D→B, D→C); none moved to a higher alternative (Figure 5). This directional homogeneity practically rules out the randomness hypothesis.

Figure 5. Item-by-item change pattern - participant P5 (anomalous case)
All 5 changes were downward (E→D, E→C, D→C, D→B, D→C)

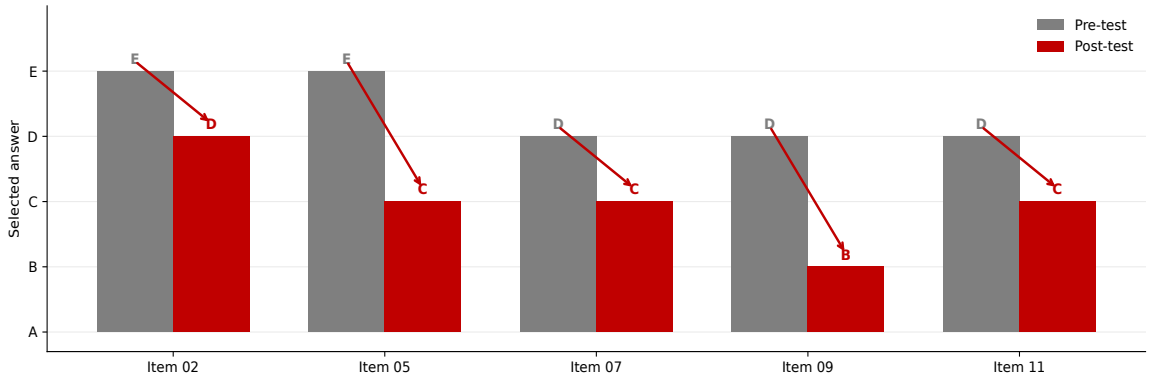


Figure 5. Item-by-item change pattern in participant P5-all five changes occurred in a downward direction.

Four interpretive hypotheses are possible. The first, genuine moral regression, is not very plausible: P5 did not move into the Primitive Band proper, but toward a darker self-assessment-a profile compatible with the Sensation/Pre-Development Band. The second, an intervening life event, cannot be ruled out without a follow-up interview, but the participant's qualitative account does not suggest an acute rupture. The third hypothesis, more clinically plausible in light of the Pyramid, is increased self-awareness: after five weeks of exposure to a more nuanced vocabulary of virtues, P5 began to see herself more clearly. At pre-test she answered as the "better" version of herself; at post-test, as the real version. The fourth hypothesis, complementary to the third, is reduced social desirability bias: the pre-test, administered at an early stage of contact with the facilitators, captured responses that P5 assumed were expected; the post-test, after a bond had been established, captured a more honest self-assessment.

The decline as evidence of the Partial Awareness Band

The interpretation most consistent with the chapter on the Pyramid of Virtues is a transition from the Pre-Development Band to the Partial Awareness Band. In this transition, pride ceases to operate as a filter of self-description, and the subject begins to tolerate seeing their own mixture without being destroyed by it. In terms of Casarotto's Law, this is exactly the kind of movement that would be expected from a successful intervention with a personality profile entering the therapeutic cycle while still strongly invested in maintaining a self-image. The uniform direction-five changes, all downward-is, from a statistical standpoint, highly unlikely under the hypothesis of randomness or measurement error. More likely, it indicates a systematic reorganization of the internal criterion of self-judgment. Virtology would call this the beginning of the dismantling of the Persona-a necessary step, although a clinically uncomfortable one.

4.6. Effect on secondary virtues: network inscription

A theoretically important finding is that the virtues showing the greatest average increases in the intervention group were not only the four trained primary virtues, but also several secondary virtues intertwined with them. The twelve secondary virtues with the greatest mean gains are presented in Figure 6.

Figure 6. Secondary virtues with the highest mean gain in the intervention group (not directly trained)

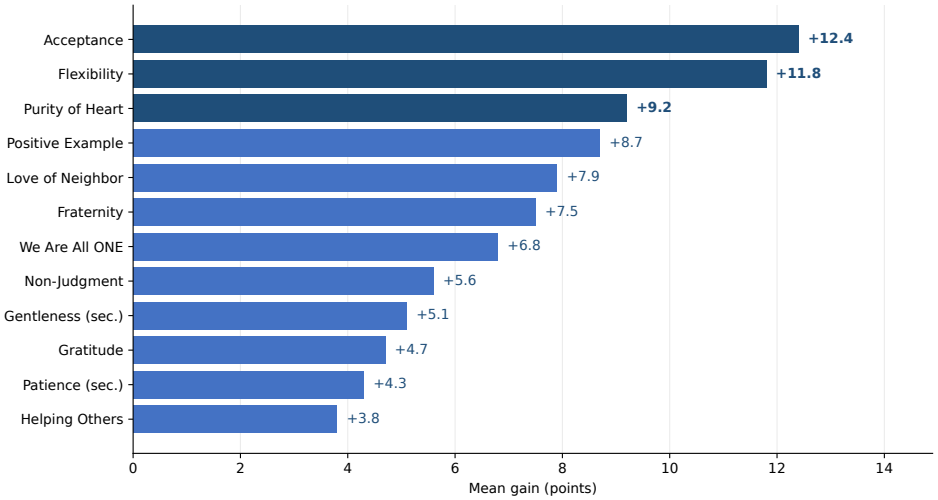


Figure 6. Secondary virtues with the greatest mean gain in the intervention group-none of which was directly trained.

Acceptance, Flexibility, Purity of Heart, Positive Example, Love of Neighbor, Fraternity, and We Are All ONE appear among the secondary virtues with the greatest gains. This pattern is exactly what the theory predicts. As the chapter devoted to virtues as neural structures explains, a virtue is not an isolated piece-it is a network that activates other networks. Training Humility activates Acceptance because both require the Ego to withstand being contradicted without collapsing. Training Gentleness activates Purity of Heart because both operate in the interval between trigger and response. Training Patience activates Fraternity because tolerance of the other's timing is, ultimately, recognition of that person as a fellow human being. By working on the four main virtues, the intervention produced positive spillover effects across the entire neural-moral network connecting them-and this is consistent with what Virtology describes as a "wholesale Band gain," in contrast to a specific gain in a single competency.

Figure 4. Heatmap of pre→post changes by participant and virtue (intervention group)



Figure 4. Heatmap of pre→post changes by participant and primary virtue in the intervention group.

The heatmap in Figure 4 reveals two movements at a glance. On the right, P1's strongly green band-the paradigmatic case-shows a jump in all virtues simultaneously. For P2 and P3, the more homogeneous green band (between +8 and +10 in all virtues) shows a pattern of balanced gains. P4 shows more modest, yet still consistently positive, gains. P5 shows the only reddish band, with a uniform decline-a case to be interpreted, as discussed, in light of the Partial Awareness Band.

4.7. Triangulation with qualitative accounts

Three spontaneous accounts collected at the end of the intervention, with identities kept confidential, provide a phenomenological grounding for the quantitative findings. The following excerpts preserve the participants' original oral form and have been anonymized.

> **Account 01-Patience applied to the relationship with a daughter**
>
> "I learned to have more patience. When I ask my daughter to do something, I'm learning to wait for her own time. I think they're learning. I also learned that we can take a different path; I don't need the path I've followed my whole life."

In terms of Casarotto's Law, E (the daughter does not immediately do what was asked) is the same, but the participant describes a change in I-"waiting for her own time" means, in physiological language, that the amygdala no longer fires as it did before, that the prefrontal cortex is managing to insert seconds between the trigger and the response, and that the somatic marker "my daughter does not obey = intolerable humiliation" has begun to be rewritten. The last sentence-"we can take a different path; I don't need the path I've followed my whole life"-is the textual sign of a change in PS: the personality is not merely performing a new behavior, but reorganizing its identity script. In contemporary neuroscientific

language, what is observed is a remodeling of prefrontal somatic markers (DAMASIO, 2010) and a strengthening of the CEN's top-down modulation of the limbic system (OCHSNER; GROSS, 2014).

Account 02-Recovery of affective tone and Positive Example

"I learned a little more patience and my feelings went back to normal; I started feeling more alive again. And I hope to change a lot and be a good mother and improve, because sometimes we just stumble through life, but we need to change to be a better influence on our children."

Two phenomena of importance to Virtology appear in this account. First, "I started feeling more alive again" describes a recovery of affective tone-in neurochemical terms, dopamine and serotonin returning to a functional level, which commonly occurs when the subject leaves a state of chronic hypervigilance (an elevated HPA axis, high cortisol, a desensitized reward system) that characterizes chronic allostatic stress (MCEWEN, 2017; SAPOLSKY, 2017). Second, the passage "I need to change to be a better influence on our children" is the textual sign of the Positive Example driver-one of the secondary virtues showing the greatest increases in the group, according to Figure 6. The participant intuitively grasped the principle articulated in the chapter devoted to this virtue: "what truly educates is not isolated discourse, but example."

Account 03-Gentleness in caring for a child with autism spectrum disorder

"I'm taking part in Virtology because we have an autistic grandson. Virtology is teaching us to deal with the problem he has. It helps us, sort of, to have more humility, not to blow up so much, right? Knowledge and knowledge to help the family."

This account is perhaps the richest clinically. "Not to blow up so much" precisely describes what the chapter on Gentleness defines as "power under command"-the prefrontal lobe holding back the explosion between the trigger and the response. In terms of contemporary neuroscience, it describes strengthened amygdala-vmPFC inhibition (ARNSTEN, 2009; OCHSNER; GROSS, 2014). For a family caring for a child with autism spectrum disorder-a condition requiring caregivers to sustain tolerance of unpredictable behaviors-the gain in Gentleness has a direct practical impact on the child's developmental environment. "Knowledge to help the family" is also the sign of Utility (G4) operating as a driving force-the virtue becomes a competency applied to the good of one's neighbor, a clinical pattern that the Pyramid associates with the transition from the Pre-Development Band to the Awareness Band.

The three accounts share a common structure when read through Casarotto's Law: E (the environmental stimulus-the daughter who does not immediately obey, the previous life condition, the autistic

grandson) remains the same, but I (decreased subjective intensity), PS (virtues incorporated as competencies), and FA (the interpretive band rises from primitive reactivity toward more elaborate configurations) change. This is precisely the operational definition of a Band gain in the virtological model-and its simultaneous presence in the quantitative and qualitative data supports the clinical interpretation of the findings.

5. Discussion

5.1. The four virtues trained in the Pyramid: crossing the Base and Middle without producing iatrogenesis

One characteristic of the intervention requires detailed theoretical analysis in light of the Pyramid of Virtues (CASAROTTO, 2026): the four virtues addressed-Humility, Gentleness, Patience, and Responsibility-do not belong to a single block of the Pyramid. Humility and Responsibility are virtues of the Base Block (Treatment), whose clinical function is to dismantle primitive structures-pride and the outsourcing of agency, respectively. Gentleness and Patience are virtues of the Middle Block (Development), associated with affective modulation and tolerance of the timing of reality. The intervention therefore crossed the first two blocks of the Pyramid simultaneously (Figure 7).

Figure 7. Position of the four trained virtues in the Pyramid of Virtues (★ = virtue trained in the intervention)

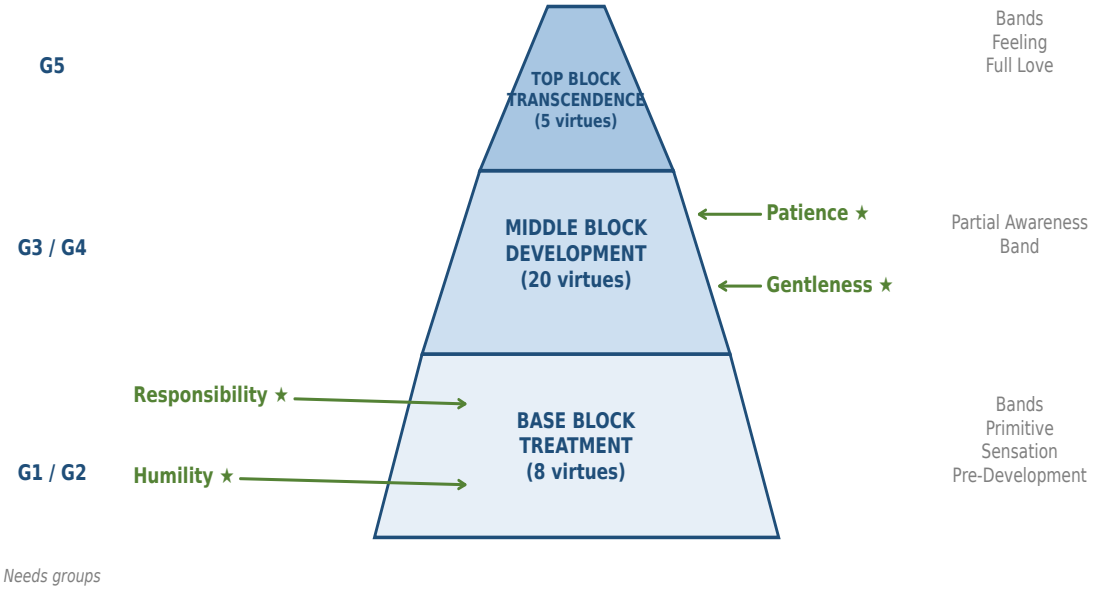


Figure 7. Position of the four trained virtues in the Pyramid of Virtues. The intervention simultaneously crossed the Base Block (Humility, Responsibility) and the Middle Block (Gentleness, Patience).

The chapter devoted to the Pyramid is explicit about the risk of "skipping stages": "Asking for charity from someone who has not yet built an Ego is the most common form of iatrogenesis in therapies inspired by virtues." It is therefore necessary to assess whether, by crossing the Base and Middle simultaneously, the intervention produced signs of virtological iatrogenesis-frustration, Persona performance, or clinical regression. The empirical analysis of the results suggests that it did not. The gains were homogeneous and balanced across the four virtues (32.5% to 37.6%), a pattern incompatible with the superficial performance of a middle virtue on an unconsolidated base-in which case one would expect a disproportionate gain in Gentleness and Patience without an accompanying gain in Humility. Furthermore, the paradigmatic case P1, starting from the lowest base, was precisely the participant with the greatest increase in Humility (from 7.0 to 30.0, more than fourfold)-the base was not skipped; it was effectively worked on and consolidated. The secondary gains in Acceptance (+12.4), Flexibility (+11.8), and Purity of Heart (+9.2)-all virtues that, according to the Pyramid, depend on prior Humility in order to become inscribed-further support this interpretation.

The interpretation most consistent with the theory is that, by combining a virtue of the Base (Humility) with a virtue of the Middle (Gentleness) that neurally complements it, and doing the same with Responsibility (Base) and Patience (Middle), the intervention operates through an architecture of pairs-each pair simultaneously covers the vector to be dismantled (pride, outsourcing) and the competency to be installed (affective modulation, tolerance of timing). This is not "skipping stages" in the sense prohibited by the Pyramid; it is "working in pairs of precondition and development," which constitutes an operational principle compatible with the clinical interpretation of "everything aligned"-virtue × need × mandate × band.

5.2. Interpretation in light of contemporary neuroscience

The findings provide preliminary evidence consistent with five converging lines of contemporary neuroscience. First, the inscription of gains within five weeks is compatible with the time window documented in the contemplative neuroscience literature: Hölzel et al. (2011) demonstrated detectable structural changes in eight weeks of MBSR; Tang et al. demonstrated functional changes in a few days of IBMT (TANG; HÖLZEL; POSNER, 2015). The virtological intervention is not equivalent to these protocols, but operates within a window compatible with what is known about trainable neural plasticity.

Second, the pattern of networked gains-secondary virtues increasing along with the trained primary virtues-is consistent with what Lazar et al. (2005) and Davidson and Begley (2012) describe about contemplative training: neural plasticity operates in networks, not isolated modules, and training a specific competency produces functional reorganization that extends to related circuits. The topography of the finding-Acceptance, Purity of Heart, Flexibility-corresponds exactly to what would be expected if Humility and Gentleness were neurally activating the same prefrontal circuits that these secondary virtues also recruit.

Third, case P1-a subject starting from the Primitive Band and moving toward Partial Awareness configurations in five weeks-is compatible with the Kandelian thesis that consolidated synaptic patterns retain residual plasticity throughout adult life, even after decades of operating within a neurotic pattern (KANDEL, 2018). The intervention acted precisely on this residual plasticity, demonstrating at the individual level what the literature demonstrates at the population level: no synaptic pattern is absolutely fixed.

Fourth, inscription in populations experiencing social vulnerability-presumably operating under a high chronic allostatic load (MCEWEN, 2017; SAPOLSKY, 2017)-suggests that the intervention can restore prefrontal function even in systems previously compromised by prolonged exposure to stress. This finding has relevant practical implications for community mental health programs: the prefrontal cortex is not merely "damaged" by stress-it is in a state of neuroplasticity that can still be mobilized, and work with virtues appears to be an effective behavioral pathway for activating it.

Fifth, case P5-increased self-awareness and reduced social desirability bias-is compatible with what Damasio (2010) describes as the reorganization of the prefrontal somatic markers that sustain the self-narrative. When the vmPFC begins to operate with less defensive criteria, the representation of the self becomes clearer-which clinically is often accompanied by a temporary decline in mood (the sadness of Awareness) and a deeper reorganization in the following months.

5.3. Network inscription and "everything aligned"

The principle of "everything aligned"-articulated in the Pyramid as a condition of effective neuroplasticity-holds that the virtue, the Group's need, the life-stage mandate, and the concrete theme of existence must be addressed simultaneously, in the same exercise, over the same period. The intervention in this study operated precisely on this principle: the exercises between meetings were paired with concrete everyday situations experienced by the participants (the daughter who does not obey, the autistic grandson, the burden of caregiving), connecting virtues of the Base and Middle with the needs of Groups 1, 2, 3, and 4 that were out of balance in real life. The pattern of networked gains emerging in the results-secondary virtues increasing in correlation with the primary virtues-is precisely what the PRF formula predicts when its factors multiply in alignment.

5.4. Why the anomalous case matters to the clinical thesis

Case P5 is not an obstacle to the findings; on the contrary, it provides indirect evidence in favor of the clinical thesis. In a merely performative intervention-in which participants only learn to mark the "correct" response-a uniform directional decline would not be observed. Any participant who had not "performed" the improvement would show either no change or a small random fluctuation. The uniform direction of P5's decline-five changes, all downward, with none upward-indicates that the

intervention produced, in that participant, an internal reorganization of the criterion of self-judgment. In virtological language: P5 did not regress; she woke up. In neuroscientific language: P5's vmPFC began to operate with less defensive somatic markers. In the language of the Pyramid: P5 began the transition between Pre-Development and Partial Awareness, a transition whose first phase is classically more uncomfortable.

Clinical implication of case P5

Virtue-based intervention programs need to be prepared for the phenomenon of "lucid decline"-participants whose scores worsen between pre- and post-test because they have begun to see themselves more clearly. Without anticipating this, the evaluator interprets the finding as a failure of the intervention when, from the Pyramid's perspective, it is a sign of partial success: the Persona has begun to give way. Clinical monitoring of these cases, with a follow-up interview and support in metabolizing emerging Awareness, is ethically necessary.

6. Limitations

The findings must be read in light of methodological limitations that should be stated. The first and most relevant is the absence of a complete post-test for the control group, for operational reasons. Without the second measurement point in the control group, it is not possible to formally attribute the intervention group's gains to the intervention within a classical experimental design-strictly speaking, one can only state that the gains occurred in the group that received the intervention, without strict causal inference. The second limitation is the small sample size (n=6 in the intervention group; n=8 in the control group), which precludes robust statistical inferential analysis and requires the findings to be treated as exploratory. The third limitation is the short time window between pre- and post-test (approximately four weeks), which prevents any claim that the gains persist beyond the immediate window-a question that only a follow-up measurement at six to twelve months could answer. The fourth limitation concerns the instrument: five virtues in the matrix (Honesty, Eternity, Detachment, Care for the Body, Love of Nature) were not weighted in the present questionnaire and could not be scored, resulting in incomplete sampling of the virtological matrix of thirty-three virtues. The fifth limitation is social desirability bias, particularly relevant in self-reported measures of virtues-a bias that the analysis of case P5 suggests was partly captured by the present instrument, but which requires triangulation with external behavioral measures in the next cycle.

Other, less relevant limitations include: internal heterogeneity in the intervention group in terms of education and family configuration; a potential Hawthorne effect (study participants tend to improve simply because they feel observed); lack of blinding of the facilitators to the study hypotheses; and a possible effect of

regression to the mean in participants who started with extreme scores, although the magnitude of P1's gains exceeds what this effect would explain.

7. Conclusions and future directions

Considering the three sources together-numbers, the facilitators' report, and spontaneous accounts-and respecting the limitations in Section 6, this study allows four central propositions to be stated, revisiting the theoretical architecture of clinical Virtology and placing the empirical findings in dialogue with it.

Proposition I-Virtues can be trained in community settings, even in populations experiencing vulnerability

A brief intervention (five meetings) based on Guia das Virtudes, conducted with systematic practice between meetings, was able to produce gains of approximately one-third of the previous score in four primary virtues, in a small sample of family members experiencing social vulnerability. These gains were accompanied by proportional increases in intertwined secondary virtues, a pattern consistent with a reorganization of the Band as a whole. The accounts confirm that the gains are not merely verbal-they describe concrete changes in everyday situations (patience with a daughter, recovery of affective tone, affective modulation in caring for an autistic child) that correspond, in the language of Casarotto's Law, to coordinated shifts in I, PS, and FA even with E constant.

Proposition II-The Pyramid of Virtues operates as a valid clinical organizer, and crossing the Base and Middle in precondition pairs does not produce iatrogenesis

The intervention simultaneously crossed the Base Block (Humility, Responsibility) and the Middle Block (Gentleness, Patience), and did not produce the signs that the Pyramid chapter associates with the iatrogenesis of "skipping stages." The gains were homogeneous and balanced, and the paradigmatic case P1 showed that Humility-the core virtue of the Base-was effectively addressed, not skipped. The most consistent interpretation is that working in pairs (a virtue of the Base with one of the Middle that neurally complements it) constitutes an operational principle compatible with the Pyramid, and warrants further theoretical development.

Proposition III-The anomalous case provides indirect evidence in favor of the clinical thesis

The uniform direction of P5's decline-five changes, all downward, with none upward-is not a failure of the intervention. In light of the Pyramid, it marks the transition from the Pre-Development Band to the Partial Awareness Band, in which pride ceases to filter self-description and the subject begins to tolerate seeing their own mixture. In neuroscientific language, it is a reorganization of prefrontal somatic markers. Future programs need to anticipate and accommodate this phenomenon-interpreting it as failure is a clinical misunderstanding.

Proposition IV-Network inscription confirms the thesis that virtues are interconnected neural structures

The secondary gain in Acceptance, Flexibility, Purity of Heart, Positive Example, Love of Neighbor, Fraternity, and We Are All ONE-virtues not directly trained-is preliminary empirical evidence for the central thesis of clinical Virtology: virtues are neural structures interconnected in a network, and directed work on four of them produces reorganization that propagates through the entire related neural-moral system. This finding connects directly with the contemporary contemplative neuroscience literature (LAZAR et al., 2005; HÖLZEL et al., 2011; TANG; HÖLZEL; POSNER, 2015; DAVIDSON; BEGLEY, 2012) and supports the virtological program as an integrating matrix for these contributions.

Directions for the next research cycle

Protocol for the next cycle of community Virtology research

1. Ensure complete administration of the post-test in the control group, even with participants partly different from those in the pre-test-the second measurement point in the control group is essential for causal inference.
2. Increase the sample size to at least twenty participants per group, allowing statistical inferential analysis (paired hypothesis tests, linear models with covariates).
3. Include follow-up measurements at six and twelve months to assess the persistence of gains-a central question for Virtology's neuroplasticity model.
4. Complete the weighting of the five virtues currently unweighted in the instrument (Honesty, Eternity, Detachment, Care for the Body, Love of Nature), allowing complete sampling of the virtological matrix of thirty-three virtues.
5. Include social desirability scale items to explicitly distinguish, in the statistical analysis, gains in virtue from gains in the ability to give the expected response.
6. Consider an instrument based on narrative vignettes (concrete scenarios with behavioral choices), which reduces socially desirable response bias and better captures the operative Evolution Band.
7. Conduct follow-up interviews with anomalous cases (P5 profiles), under the hypothesis of a transition between Bands, and with a protocol for clinically supporting emerging Awareness.
8. Triangulate self-reported measures with external observational measures-accounts from family members living with the participants, direct observation in structured situations, and physiological markers when possible (heart rate variability as a proxy for vagal modulation and prefrontal function).

Strictly speaking, the research presented here is the first step in a program for the empirical validation of clinical Virtology in a naturalistic setting. The findings support, with epistemological humility, that the path is fruitful-virtues can be trained, the Pyramid operates as an

organizer, network inscription is detectable, and the anomalous case can be interpreted within the theoretical framework itself. The program's next stages require methodological refinement, larger samples, and longitudinal measures. This article is the starting point, not the conclusion; but it is a starting point at which empirical investigation begins to confirm, in a real-world setting, what the theory has been articulating for years.

> ***The pyramid is the map, and the sequence is the path. Each participant climbs one step at a time-humility before acceptance, acceptance before gentleness, gentleness before love of neighbor. Skipping steps means leaving the participant where they were, dressed in the Persona of someone who has arrived. The findings presented here suggest, on a preliminary empirical scale, that the sequence works.***

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